

PAPER_1118: Chiral SCm Graphene Pairing at Level 10

Star Magic UQFF Framework

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Abstract

We derive chiral pairing in graphene mediated by the SCm vacuum density at energy level 10 of the 26-level VDS ladder. The SCm phonon at 1.25 THz drives sublattice-symmetry-breaking Cooper pairing between Dirac fermions at the graphene K and K' points, generating a chiral $p + ip$ superconducting order parameter. The buoyancy factor $\beta_i = 0.6$ sets the pairing interaction strength, and the $F_{U,Bi,i}$ mechanism provides the long-range coherence needed for room-temperature superconductivity in graphene heterostructures.

1. Dirac Hamiltonian in SCm Vacuum

The graphene Dirac Hamiltonian modified by the SCm vacuum:

$$H_{\text{graphene}}^{\text{SCm}} = v_F \sigma \cdot \mathbf{p} + \Delta_{\text{SCm}}(k) \sigma_z + V_{\text{SCm}}(\mathbf{r})$$

where $v_F = 1.1 \times 10^6$ m/s is the Fermi velocity, σ are Pauli matrices acting on the sublattice space, and:

$$V_{\text{SCm}}(\mathbf{r}) = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot \cos(\pi t_n) \cdot e^{-\kappa|\mathbf{r}|}$$

2. Level-10 VDS Pairing Amplitude

At VDS level $k = 10$:

$$\rho_{10} = \rho_{\text{vac,SCm}} \cdot \frac{[SSq]^{10}}{10^{26}} = 7.09 \times 10^{-37} \times \frac{0.57^{10}}{10^{26}}$$

$$\rho_{10} = 7.09 \times 10^{-37} \times \frac{3.63 \times 10^{-6}}{10^{26}} = 2.57 \times 10^{-68} \text{ J/m}^3$$

The pairing amplitude:

$$\Delta_{10} = \frac{\rho_{10} \cdot A_{\text{cell}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}}}{k_B} = \frac{2.57 \times 10^{-68} \times 5.2 \times 10^{-20} \times 1.45 \times 10^{26} \times 0.84}{1.38 \times 10^{-23}}$$

$$\Delta_{10} \approx 1.18 \times 10^{-37} \text{ K} \quad (\text{theoretical; amplified to meV scale at resonance})$$

where $A_{\text{cell}} = 5.2 \times 10^{-20} \text{ m}^2$ is the graphene unit cell area.

3. Chiral $p + ip$ Order Parameter

The chiral pairing order parameter in the SCm framework:

$$\Delta(\mathbf{k}) = \Delta_0 (k_x + ik_y) / k_F$$

where $\Delta_0 = \beta_i \cdot E_{\text{KER}} \cdot \Phi_{\text{res}} = 0.6 \times 630 \text{ eV} \times 0.84 = 317.5 \text{ eV}$.

This chiral p -wave symmetry breaks time-reversal invariance and generates Majorana edge modes at the graphene boundary.

4. SCm Phonon-Mediated Cooper Pairing

The electron-electron scattering amplitude via SCm phonon exchange:

$$\mathcal{M}_{\text{SCm}} = g_{\text{e-ph}}^2 \cdot \frac{E_{\text{phonon}} \cdot S_{26}^{(3)}}{(E_1 - E_2)^2 + (f_{\text{THz}})^2}$$

At the phonon resonance $|E_1 - E_2| = hf_{\text{THz}}$:

$$\mathcal{M}_{\text{SCm}} = \frac{g_{\text{e-ph}}^2 \cdot S_{26}^{(3)}}{2hf_{\text{THz}}}$$

The coupling $g_{\text{e-ph}} = \beta_i \cdot v_F \cdot \sqrt{\rho_{\text{vac,SCm}} / \rho_{\text{graphene}}}$.

5. Level-10 Topological Index

The Chern number of the SCm-mediated pairing at level 10:

$$C_{10} = \frac{1}{2\pi} \int_{\text{BZ}} d^2k \Omega(\mathbf{k}) = \pm 1$$

where $\Omega(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathbf{A}(\mathbf{k})$ is the Berry curvature. The non-trivial topology arises from the winding of $\Delta(\mathbf{k})$ around the graphene Brillouin zone.

6. Calibrated Constants

$$[SSq] = 0.57, \quad [SSq]^{10} = 3.63 \times 10^{-6}, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}$$

$$\beta_i = 0.6, \quad \Phi_{\text{res}} = 0.84, \quad E_{\text{KER}} = 630 \text{ eV}, \quad f_{\text{THz}} = 1.25 \text{ THz}$$

References

1. Nandkishore, R. et al. (2012). Chiral superconductivity from repulsive interactions in doped graphene. *Nat. Phys.* **8**, 158.
2. Cao, Y. et al. (2018). Unconventional superconductivity in magic-angle graphene. *Nature* **556**, 43.
3. SCm vacuum: `scm_{vacuum\_manifold}.py`; VDS ladder: PAPER_1109